A Smooth Recurrent-Spike Obstruction for Weighted Boundary Embeddings of Dirichlet Series

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Status: candidate partial result. The source asks which nonnegative weights ω yield a bounded critical-line embedding for the Hardy–Hilbert space of Dirichlet series. We prove that the necessary condition $\omega \in L^1 \cap L^\infty$ is not sufficient, even for smooth weights tending to zero. In fact, a bad weight can have an admissible symmetric decreasing rearrangement.

1 Source problem and result

Let $\mathscr{H}^2$ be the Hilbert space of Dirichlet series

$$f(s) = \sum_{n\geq 1} a_n n^{-s}, \qquad \|f\|^2_{\mathscr{H}^2} = \sum_{n\geq 1} |a_n|^2.$$

Brevig and Perfekt ask which nonnegative measurable ω satisfy

$$\int_{\mathbb{R}} |f(1/2+it)|^2 \omega(t)\, dt \leq C_\omega \|f\|^2_{\mathscr{H}^2} \qquad (f \in \mathscr{H}^2). \tag{1}$$

They note that $L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$ is necessary and prove (1) when $\omega(t) = v(|t|)$ with v decreasing and $v \in L^1 \cap L^\infty$; see Lemma 7.2 and the ensuing open problem in arXiv:2005.05094 [1].

Theorem 1. *There is a nonnegative function $\omega \in C^\infty(\mathbb{R})$ such that*

$$\lim_{|t|\to\infty} \omega(t) = 0, \qquad \omega \in L^q(\mathbb{R}) \quad (1 \leq q \leq \infty),$$

but (1) *fails. Moreover, the symmetric decreasing rearrangement $\omega^\#$ of ω satisfies* (1)*. Consequently admissibility is not rearrangement invariant; in particular, it cannot be characterized by the distribution function of ω or by rearrangement-invariant norms alone.*

2 Proof intuition

The normalized truncated zeta kernel

$$F_N(s) = \frac{1}{\sqrt{H_N}} \sum_{n\leq N} \frac{1}{\sqrt{n}} n^{-s}, \qquad H_N = \sum_{n\leq N} \frac{1}{n},$$

has $\mathscr{H}^2$ norm one. On the critical line it has height about $\sqrt{\log N}$ on a window of width about $1/\log N$ around $t = 0$. The logarithms of the primes are rationally independent, so Kronecker's theorem makes all phases n^{-it}, $n \le N$, return simultaneously near 1 at arbitrarily many widely separated times. Hence the same narrow, high peak recurs as often as desired.

Place smooth bumps of height h_k on K_k such recurrences for a rapidly growing cutoff N_k. Their total L^1 cost is of order $h_k K_k / \log N_k$, while testing against F_{N_k} yields a contribution of order $h_k K_k$. Choosing $h_k = 1/k$, $K_k = k^2$, and N_k sufficiently large makes the first quantities summable and the second tend to infinity.

3 The recurrent peak

Lemma 2. *Fix $N \ge 2$. There are arbitrarily large positive numbers τ such that, whenever $|u| \le (16 \log N)^{-1}$,*

$$|F_N(1/2 + i(\tau + u))|^2 \ge \frac{49}{64} H_N.$$

Consequently one may choose any prescribed finite number of such τ, with the corresponding intervals pairwise disjoint and lying beyond any previously fixed compact set.

Proof. Let $\mathcal{P}_N$ be the finite set of primes at most N, and put $r_N = \lfloor \log_2 N \rfloor$. Unique factorization shows that $\{\log p : p \in \mathcal{P}_N\}$ is linearly independent over $\mathbb{Q}$. Kronecker's theorem therefore gives arbitrarily large $\tau > 0$ for which

$$|p^{-i\tau} - 1| \le \frac{1}{16 r_N} \qquad (p \in \mathcal{P}_N).$$

Writing $n \le N$ as a product of at most r_N primes, counted with multiplicity, and using $|z_1 \cdots z_m - 1| \le \sum_j |z_j - 1|$ for unimodular z_j, gives $|n^{-i\tau} - 1| \le 1/16$. If $|u| \le (16 \log N)^{-1}$, then

$$|n^{-iu} - 1| \le |u| \log n \le \frac{1}{16},$$

and hence $|n^{-i(\tau+u)} - 1| \le 1/8$ for every $n \le N$. It follows that

$$\begin{aligned} \left| \sum_{n \le N} n^{-1-i(\tau+u)} \right| &\ge H_N - \sum_{n \le N} \frac{|n^{-i(\tau+u)} - 1|}{n} \\ &\ge \frac{7}{8} H_N. \end{aligned}$$

Division by $\sqrt{H_N}$ proves the estimate. The positive tail of the dense Kronecker orbit is dense as well, so the returns can be chosen successively beyond arbitrary compact sets. □

4 Construction of the weight

Choose integers $N_k \ge 2$ so rapidly that

$$\log N_k \ge 2^k k^2, \tag{2}$$

and set

$$h_k = \frac{1}{k}, \qquad K_k = k^2, \qquad \delta_k = \frac{1}{16 \log N_k}.$$

Using Lemma 2, choose K_k recurrence centers $\tau_{k,1}, \dots, \tau_{k,K_k}$ so that every interval $(\tau_{k,j} - \delta_k, \tau_{k,j} + \delta_k)$ is disjoint from all the others, and arrange the blocks successively toward $+\infty$.

Let $\eta \in C_c^\infty((-1,1))$ satisfy $0 \le \eta \le 1$ and $\eta = 1$ on $[-1/2, 1/2]$, and define

$$\omega(t) = \sum_{k\ge 1} h_k \sum_{j=1}^{K_k} \eta\left(\frac{t-\tau_{k,j}}{\delta_k}\right). \tag{3}$$

The supports are disjoint and locally finite. Thus ω is smooth, $0 \le \omega \le 1$, and $\omega(t) \to 0$ as $|t| \to \infty$ because $h_k \to 0$.

For every $1 \le q < \infty$, disjointness, a change of variables, and (2) give

$$\int_{\mathbb{R}} \omega(t)^q \, dt = \|\eta\|_{L^q}^q \sum_{k\ge 1} K_k h_k^q \delta_k \le \frac{\|\eta\|_{L^q}^q}{16} \sum_{k\ge 1} \frac{k^{-q}}{2^k} < \infty.$$

Hence $\omega \in L^q$ for all stated q, including $q = \infty$.

On the core interval $|t - \tau_{k,j}| \le \delta_k/2$, the jth bump equals h_k, and Lemma 2 applies. Since $H_N \ge \log(N+1) > \log N$,

$$\begin{aligned}\int_{\mathbb{R}} |F_{N_k}(1/2+it)|^2 \omega(t)\, dt &\ge K_k h_k \delta_k \frac{49}{64} H_{N_k} \\ &\ge \frac{49}{1024} K_k h_k = \frac{49}{1024} k \longrightarrow \infty.\end{aligned}$$

But $\|F_{N_k}\|_{\mathscr{H}^2} = 1$, so (1) fails.

Finally, $\omega^\#$ is an even radially decreasing nonnegative function equimeasurable with ω. In particular it lies in $L^1 \cap L^\infty$. Lemma 7.2 of the source paper applies to $\omega^\#$ and proves its admissibility, completing the proof of Theorem 1.

5 A useful exact formulation and the remaining gap

For $\omega \in L^1(\mathbb{R})$, write $\widehat{\omega}(\xi) = \int_{\mathbb{R}} \omega(t) e^{-it\xi}\, dt$. Expanding a Dirichlet polynomial shows that (1) is equivalent to uniform boundedness on ℓ^2 of the finite sections of the positive matrix

$$G_\omega(m,n) = \frac{\widehat{\omega}(\log(m/n))}{\sqrt{mn}}. \tag{4}$$

This is an exact coefficient criterion, but it is not yet a geometric description of the weights.

The source's local embedding theorem also immediately gives the broader sufficient condition

$$\sum_{j\in\mathbb{Z}} \operatorname*{ess\,sup}_{t\in[j,j+1]} \omega(t) < \infty.$$

The construction above shows why $L^1 \cap L^\infty$ alone misses essential information: arithmetic placement of thin pieces can synchronize with simultaneous almost-periodic recurrences. A full intrinsic characterization remains open; the present theorem rules out the most natural rearrangement-invariant answer.

References

[1] O. F. Brevig and K.-M. Perfekt, *A mean counting function for Dirichlet series and compact composition operators*, Adv. Math. 385 (2021), 107775; arXiv:2005.05094.

[2] H. Hedenmalm, P. Lindqvist, and K. Seip, *A Hilbert space of Dirichlet series and systems of dilated functions in* $L^2(0,1)$, Duke Math. J. 86 (1997), 1–37.

[3] J.-F. Olsen and E. Saksman, *On the boundary behaviour of the Hardy spaces of Dirichlet series and a frame bound estimate*, J. Reine Angew. Math. 663 (2012), 33–66; arXiv:0907.2708.